

Tri-Modal Sequence Model (TSM) - Full Explanation with Math

1. Core Idea

TSM combines Transformer (attention), Mamba (state memory), and Hyena (pattern filtering) into a unified architecture.

2. Representation Mismatch

Each model produces different representations. Transformer encodes relationships, Mamba encodes state memory, and Hyena encodes patterns. These are not directly compatible.

3. Projection (Alignment Step)

To align representations: $T' = W_T T$ $M' = W_M M$ $H' = W_H H$ This maps all outputs into a shared space of dimension d .

4. Normalization

Normalize outputs to prevent dominance: $\tilde{T} = \text{LayerNorm}(T')$ $\tilde{M} = \text{LayerNorm}(M')$ $\tilde{H} = \text{LayerNorm}(H')$

5. Adaptive Gating

Compute importance weights: $g = \text{softmax}(W_g [\tilde{T} \parallel \tilde{M} \parallel \tilde{H}])$ $g = [g_T, g_M, g_H]$

6. Fusion Equation

Final unified representation: $U = g_T * \tilde{T} + g_M * \tilde{M} + g_H * \tilde{H}$

7. Alignment Loss (optional)

To force representations to align: $L_{\text{align}} = ||T' - M'||^2 + ||M' - H'||^2$

8. Key Challenges

Representation mismatch, module dominance, training instability, and fusion complexity.

9. Insight

Fusion is not simple addition, but dynamic selection of intelligence sources.